

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I SAIRAM 192424111 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

Scenario:

A company is transmitting repeated financial data blocks over a network. Initially, they used Electronic Codebook (ECB) mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

A block cipher encrypts data in fixed-size blocks. In ECB (Electronic Codebook) mode, each plaintext block is encrypted independently using the same secret key.

Encryption Formula:

$$C_i = E(K, P_i)$$

Where:

P_i = Plaintext block

C_i = Ciphertext block

K = Secret key

Since every block is encrypted independently with the same key, identical plaintext blocks always produce identical ciphertext blocks.

For example,

Plaintext:

Salary = ₹50,000

Salary = ₹50,000

Salary = ₹50,000

ECB Encryption:

Block 1 → A7F92

Block 2 → A7F92

Block 3 → A7F92

The repeated ciphertext clearly reveals that the plaintext blocks are identical. Although the attacker cannot read the actual message, they can identify patterns in the encrypted data.

Disadvantages of ECB

- Identical plaintext produces identical ciphertext.
- Reveals data patterns.
- Vulnerable to statistical analysis.
- Not suitable for financial or confidential information.
- Does not provide semantic security.

Hence, ECB mode is not recommended for transmitting sensitive financial data because it leaks information through repeated ciphertext patterns.

B. Suggest a more secure block cipher mode and justify your choice.

The most suitable mode is CBC (Cipher Block Chaining).

In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption. The first block uses an Initialization Vector (IV).

Encryption Process:

First Block

$P1 \text{ XOR } IV \rightarrow \text{Encrypt} \rightarrow C1$

Second Block

$P2 \text{ XOR } C1 \rightarrow \text{Encrypt} \rightarrow C2$

Third Block

$P3 \text{ XOR } C2 \rightarrow \text{Encrypt} \rightarrow C3$

Even if two plaintext blocks are identical,

$P1 = P2 = P3$

their ciphertext becomes

$C1 \neq C2 \neq C3$

because every block depends on the previous ciphertext block.

Advantages of CBC

- Uses an Initialization Vector (IV).
- Eliminates repeated ciphertext patterns.
- Provides better confidentiality.
- Resistant to pattern analysis.
- Suitable for banking and financial applications.
- More secure than ECB for transmitting sensitive information.

Conclusion

CBC mode should be used instead of ECB because it hides repeated patterns by chaining every block with the previous ciphertext block. This provides much stronger security for transmitting repeated financial data over a network.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

B. Justify your choice based on security strength and efficiency.

Scenario:

A banking application requires high security and moderate performance for encrypting customer transactions.

Available algorithms:

- DES
- 3-DES
- Blowfish

A. Identify the most suitable algorithm.

Answer: Blowfish

Blowfish is the most suitable algorithm because it provides a good balance between strong security and fast encryption performance.

B. Justify your choice based on security strength and efficiency.

1. DES (Data Encryption Standard)

Key Size: 56 bits

Features:

- Uses a 56-bit secret key.
- Performs 16 rounds of encryption.
- Developed in the 1970s.

Advantages:

- Simple implementation.
- Fast on older hardware.

Disadvantages:

- Small key size.
- Easily broken using brute-force attacks.
- No longer secure for modern banking systems.

Therefore, DES is not suitable for protecting customer transactions.

2. Triple DES (3-DES)

Key Size:

- 112 bits (2-key version)
- 168 bits (3-key version)

Working:

Encrypt → Decrypt → Encrypt

Advantages:

- More secure than DES.
- Resistant to brute-force attacks.
- Used in many legacy banking systems.

Disadvantages:

- Performs encryption three times.
- Much slower than DES and Blowfish.
- High computational overhead.

Although secure, 3-DES is slower and less efficient for modern banking applications.

3. Blowfish

Key Size:

32 bits to 448 bits

Features:

- Symmetric block cipher.
- 16-round Feistel network.
- Variable key length.
- Fast encryption.
- Free and unpatented.

Advantages:

- Strong security.
- Supports keys up to 448 bits.
- Faster than 3-DES.
- Difficult to break using brute-force attacks.
- Suitable for high-security applications.

Disadvantages:

- Slower key generation compared to newer algorithms like AES.
- Not ideal when keys change very frequently.

Feature	DES	3-DES	Blowfish
Key Size	56 bits	112/168 bits	32–448 bits
Security	Low	High	Very High
Speed	Fast	Slow	Fast
Brute-force Resistance	Weak	Strong	Very Strong
Encryption Process	Single encryption	Triple encryption (Encrypt–Decrypt–Encrypt)	16-round Feistel network
Performance	Good	Moderate	High
Suitable for Banking	No	Yes	Best Choice

Why Blowfish is the Best Choice

For a banking application:

- Customer information must remain confidential.
- Encryption should resist brute-force attacks.
- Transactions should be processed quickly.
- The algorithm should provide both security and efficiency.

Blowfish satisfies all these requirements because:

- It supports key sizes up to 448 bits.
- It provides stronger security than DES.
- It is faster than 3-DES.
- It offers an excellent balance between performance and security.

Conclusion

Among DES, 3-DES, and Blowfish, Blowfish is the most suitable encryption algorithm for banking systems because it provides high security, fast encryption, flexible key sizes, and efficient performance. It ensures secure customer transactions while maintaining moderate processing speed.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided

Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand
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